

Lecture 3

SOFTWARE

Operating System





LECTURE CONTENT

- Software and its types
- Operating System and its types
- Functions of OS
- OS Market share
- Dual Boot vs. Virtual Machine
- GUI and CLI

SOFTWARE

Software refers to a program or a set of instructions and applications used to manage and control various functions of a device such as a computer.

There are two main categories of computer software:

- 1) system software;
- 2) application software.



SYSTEM SOFTWARE

System software serves as a base for application software. It controls the basic (and invisible to the user) functions of a computer and comes usually preinstalled with the machine. System software includes device drivers, operating systems (OSs), compilers, text editors, and utilities helping the computer to operate more efficiently. It is also responsible for managing hardware components and providing basic non-task-specific functions.

SYSTEM SOFTWARE

- **Operating system:** harnesses communication between hardware, system programs, and other applications.
- **Device driver:** enables device communication with the OS and other programs.
- **Firmware:** enables device control and identification.
- **Translator:** translates high-level languages to low-level machine codes.
- **Utility:** ensures optimum functionality of devices and applications.

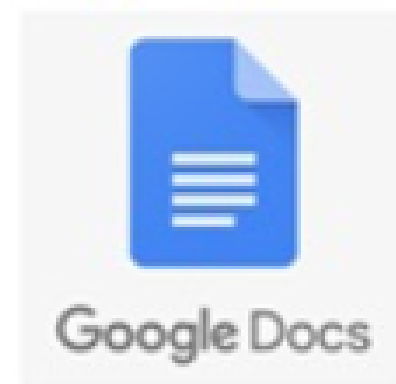
APPLICATION SOFTWARE

Application software is intended to perform specific tasks. It handles multitudes of standard and specialized tasks a user wants to achieve, such as accounting, communicating, data processing, word processing.

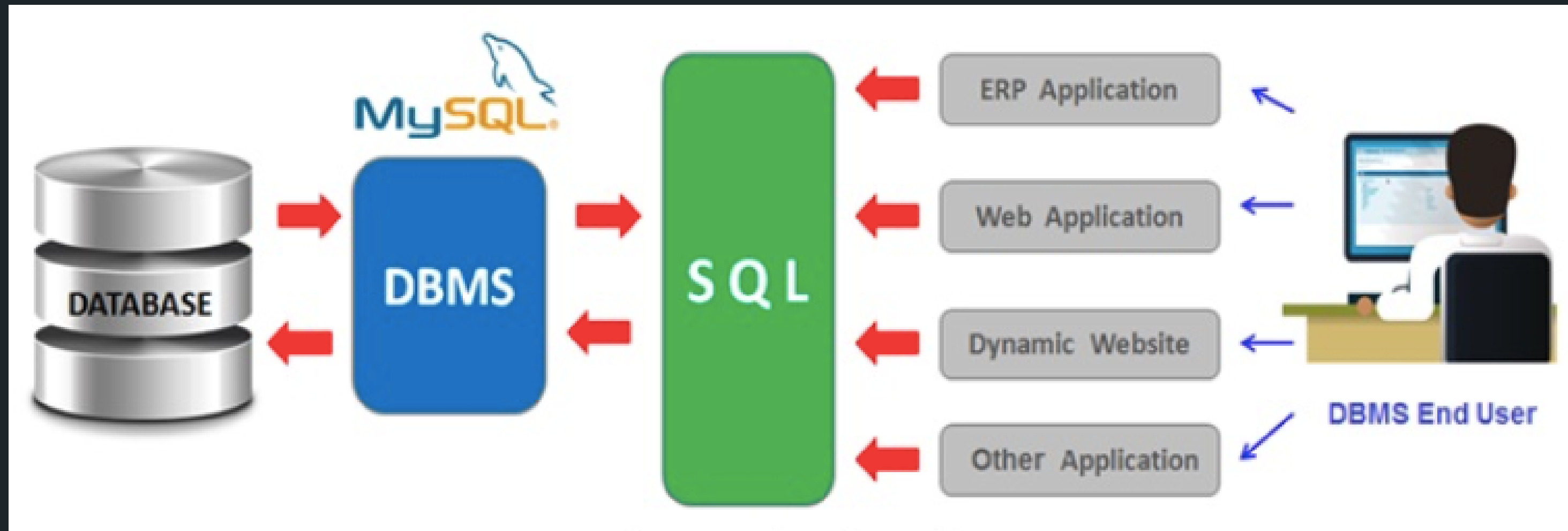
APPLICATION SOFTWARE

- **Word Processor:** MS Word, WordPad and Notepad, etc.
- **Database:** Oracle, MS Access, etc.
- **Spreadsheet:** Apple Numbers, Microsoft Excel, etc.
- **Presentation:** Microsoft Power Point, Keynotes, etc.
- **Enterprise Software:** Customer relationship management system

GENERAL PURPOSE APPLICATION SOFTWARE



CUSTOM SOFTWARE APPLICATION



Application Software can be developed as



Desktop application



Mobile application



Cloud based software applications



Web application

SOFTWARE

There are also a lot of free software programs available that are separated into the following categories.

- ***Shareware or trial software*** is software that gives you a few days to try the software before you have to buy the program. After the trial time expires, you'll be asked to enter a code or register the product before you can continue to use it.
- ***Freeware*** is completely free software that never requires payment, as long as it is not modified.
- ***Open source software*** is similar to freeware. Not only is the program free, but the source code is also available to everyone.
- ***Bundle ware*** is software bundled with new computers and hardware.

WHAT WAS THE FIRST PIECE OF COMPUTER SOFTWARE?

The first software program that was held in electronic memory was credited to and written by Tom Kilburn. The program calculated the highest factor of the integer 218 = 262,144, and was successfully executed on June 21, 1948, at the University of Manchester, England. The computer that held that program was called the SSEM (Small Scale Experimental Machine), otherwise known as the "Manchester Baby." This event is widely celebrated as the birth of software.

1947/48 - Kilburn Highest Factor Routine (amended) -

Line	C	26	26 ²	27	Line	012345	1345
-245 C	-G ₁	-	-	-	1	00011	010
-2526	-	-G ₁	-	-	2	01011	110
-265 C	G ₁	-	-	-	3	01011	010
-2727	-	-G ₁	G ₁	-	4	11011	110
-285 C	a	T ₀₁	-G ₁	G ₁	5	11101	010
Subr. 27	a-b ₁	-	-	-	6	11011	001
Subr. 26	-	-	-	-	7	-	011
Subr. 25	-	-	-	-	8	00101	100
-265 C	r ₁	r ₁	-	-	9	01011	001
-2727	-	-	-	-	10	10011	110
-285 C	-	-	-	-	11	10011	010
-2926	-	-	-	-	12	-	011
Stop	0	0	-G ₁	G ₁	13	-	111
-265 C	G ₁	r ₁	-G ₁	G ₁	14	01011	010
Subr. 21	G ₁	-	-	-	15	10101	001
-2727	G ₁	-	-	-	16	11011	110
-285 C	-G ₁	-	-	-	17	11011	010
-2926	-	-	-	-	18	01011	110
22568	r ₁	-G ₁	G ₁	-	19	01101	000

20	-3	1011125
21	1	10000
22	4	00100

or 10100

23	-2
24	G ₁

25	-	r ₁ G ₁
26	-	-G ₁
27	-	G ₁

The SSEM's first program.

An operating system is a type of software interface between the user and the device hardware. It allows users to communicate with the device and perform the desired functions.

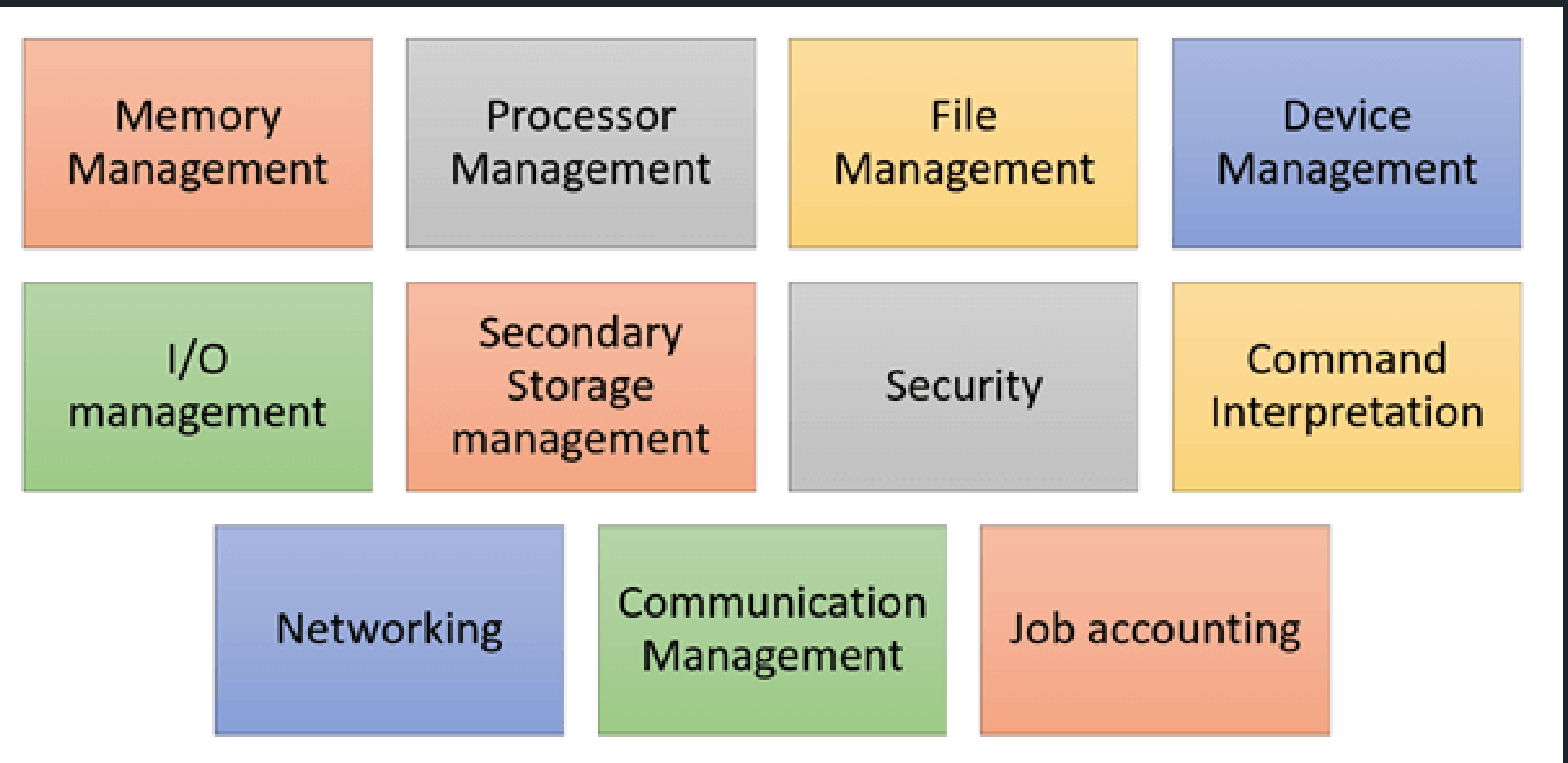


Every computer, smartphone or similar electronic device comes with special software called an operating system. An operating system, also known as an OS, is the engine behind the utility value of computers and smartphones.

- An operating system is the software that supports and manages all the programs and applications used by a computer or mobile device.
- An operating system uses a graphic user interface (GUI), a combination of graphics and text, that allows you to interact with the computer or device.
- Every computer or smart device needs at least one operating system to run applications and perform tasks.



MAIN FUNCTIONS OF OPERATING SYSTEM



TYPES OF OPERATING SYSTEM (OS)

Following are the popular types of OS:

- Batch Operating System
- Multitasking/Time Sharing OS
- Multiprocessing OS
- Real Time OS
- Distributed OS
- Network OS
- Mobile OS

WHAT IS A 32-BIT OPERATING SYSTEM?

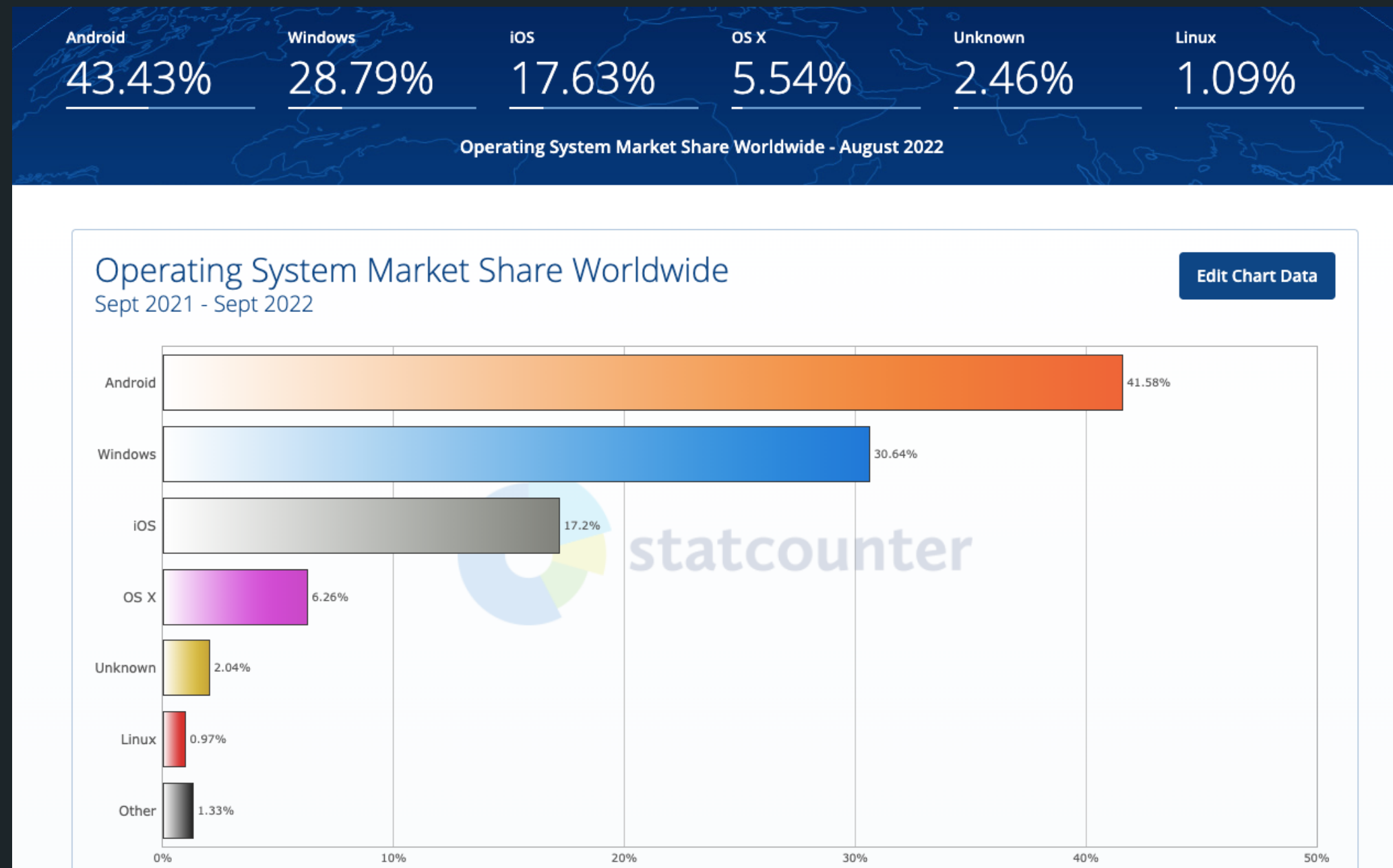
It is a CPU architecture type that holds the capacity to transfer 32 bits of data. It refers to the amount of data and information that your CPU can easily process when performing an operation. A majority of the computers produced in the early 2000s and 1990s were 32-bit machines.

WHAT IS A 64-BIT OPERATING SYSTEM?

The 64-bit microprocessor allows computer systems to process information, data, and memory addresses represented by 64 bits. Such a system can typically reference 16 exabytes (17,179,869,184 GB), or 18,446,744,073,709,551,616 bytes of memory.

A 64-bit system (a computer with a 64-bit processor) has the capacity to access more than 4 GB of RAM. It means that if a computer has 8 GB of RAM, it requires a 64-bit processor. Or else, the CPU will be inaccessible to at least 4 GB of the memory.

THE MOST WIDELY USED OS RIGHT NOW



As of September 2022, StatCounter's data for web client OS reported that the most popular operating system right now is Android, with 43.43%.

This figure is followed by Windows, and 3rd is Apple's iOS and 4th macOS. Hence, the thought of Linux as the most significant OS is also considered since Android is a modified Linux version. On the desktop space, however, most devices use Microsoft's Windows operating system.

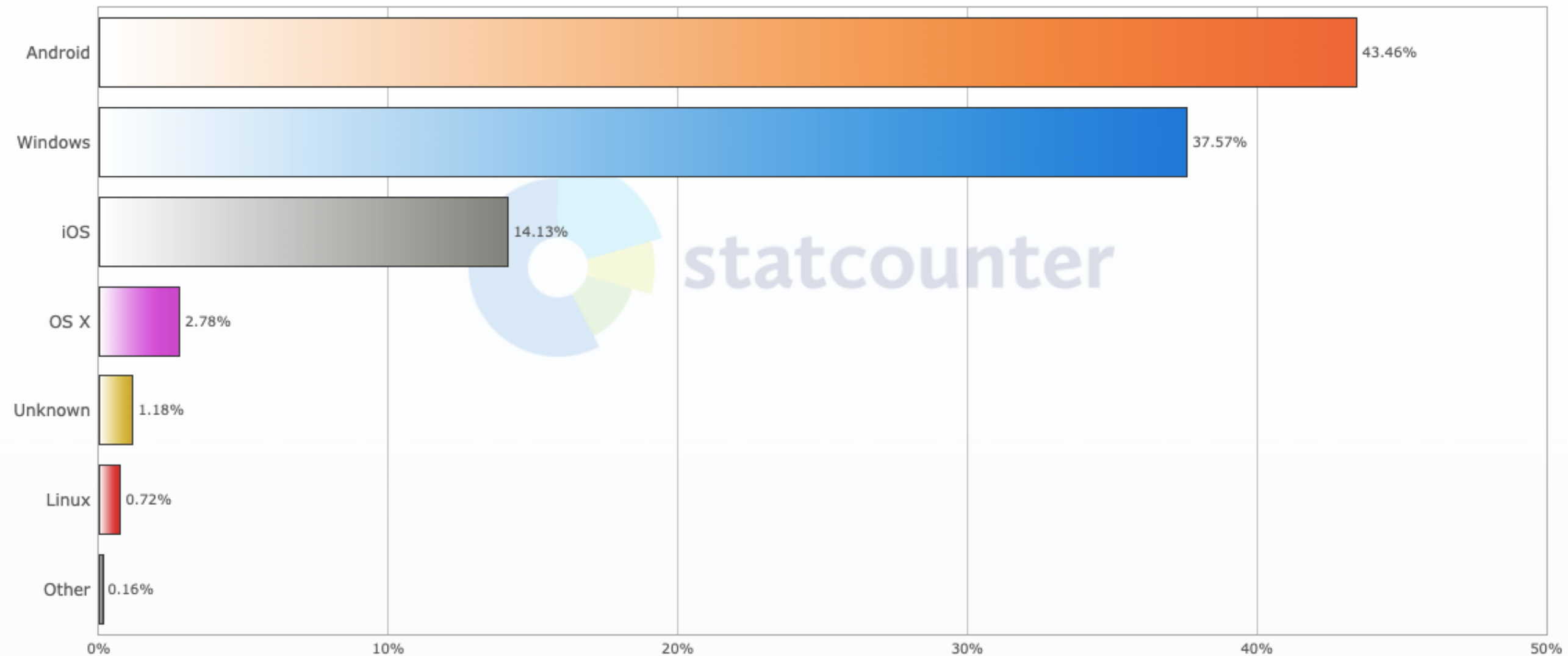
THE MOST WIDELY USED OS IN KAZAKHSTAN



Operating System Market Share Kazakhstan

Sept 2021 - Sept 2022

[Edit Chart Data](#)



DUAL BOOT VS. VIRTUAL MACHINE

If you want to run multiple operating systems on one machine, or multiple copies of the same operating system, there are two main ways to do it: dual-booting or using a virtual machine. Both methods are useful, but they serve different purposes.

Normally, 100 percent of your computer's hardware is dedicated to running your operating system. But when you use a **virtual machine**, you're running a second operating system within your primary operating system. This means you have to split your hardware between the two.

This is a problem for underpowered computers. If your computer struggles to run your primary OS smoothly, adding another one is only going to make it worse.

Suddenly dedicating half of your RAM to a secondary OS will make your primary OS chug if you're low on RAM to begin with.

DUAL BOOT VS. VIRTUAL MACHINE

On older computers and laptops, **dual-booting** is really your only option. When you dual boot, you can switch between operating systems at boot, which allows you to dedicate all your hardware to one at a time.

The more powerful your hardware, the more viable a virtual machine becomes. How much power a virtual machine requires depends on the operating system. For example, if you're virtualizing Windows 10 within macOS, then you'll need a more modern rig.

DUAL BOOT VS. VIRTUAL MACHINE

Virtual machines are great for multitasking, allowing you to switch between multiple operating systems with a simple Alt + Tab. This makes them the better choice if you need to switch between OSes regularly.

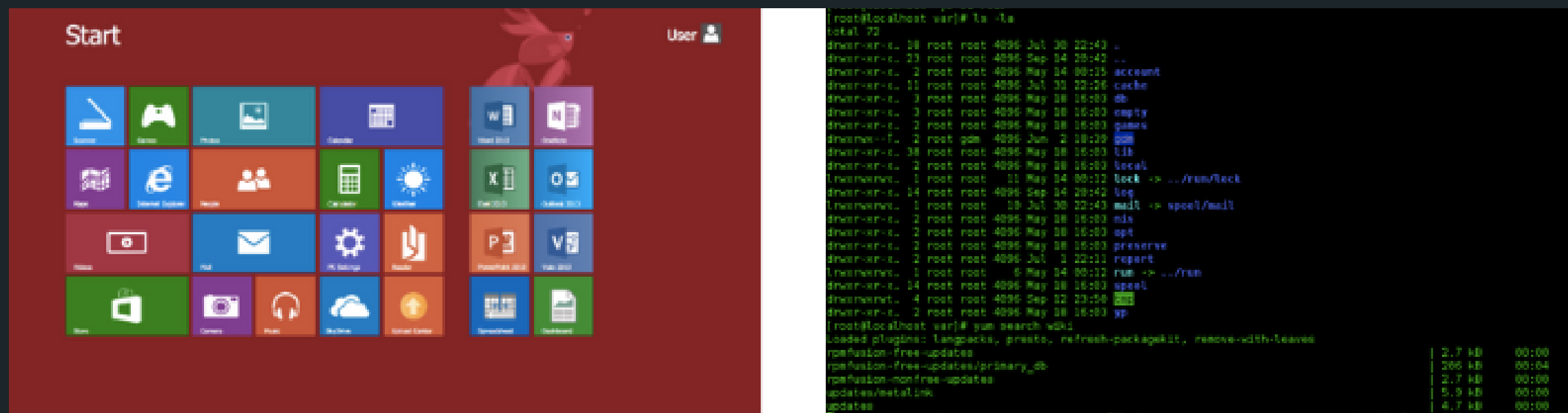
However, if you're going to work inside a virtualized operating system for hours at a time, virtual machines aren't the most efficient choice.

For example, let's say your main operating system is Windows but you're a programmer and prefer to code in a Linux environment. If you're going to bury yourself in a three-hour coding session, you might as well dual-boot into Linux and take full advantage of the speed of a native operating system.

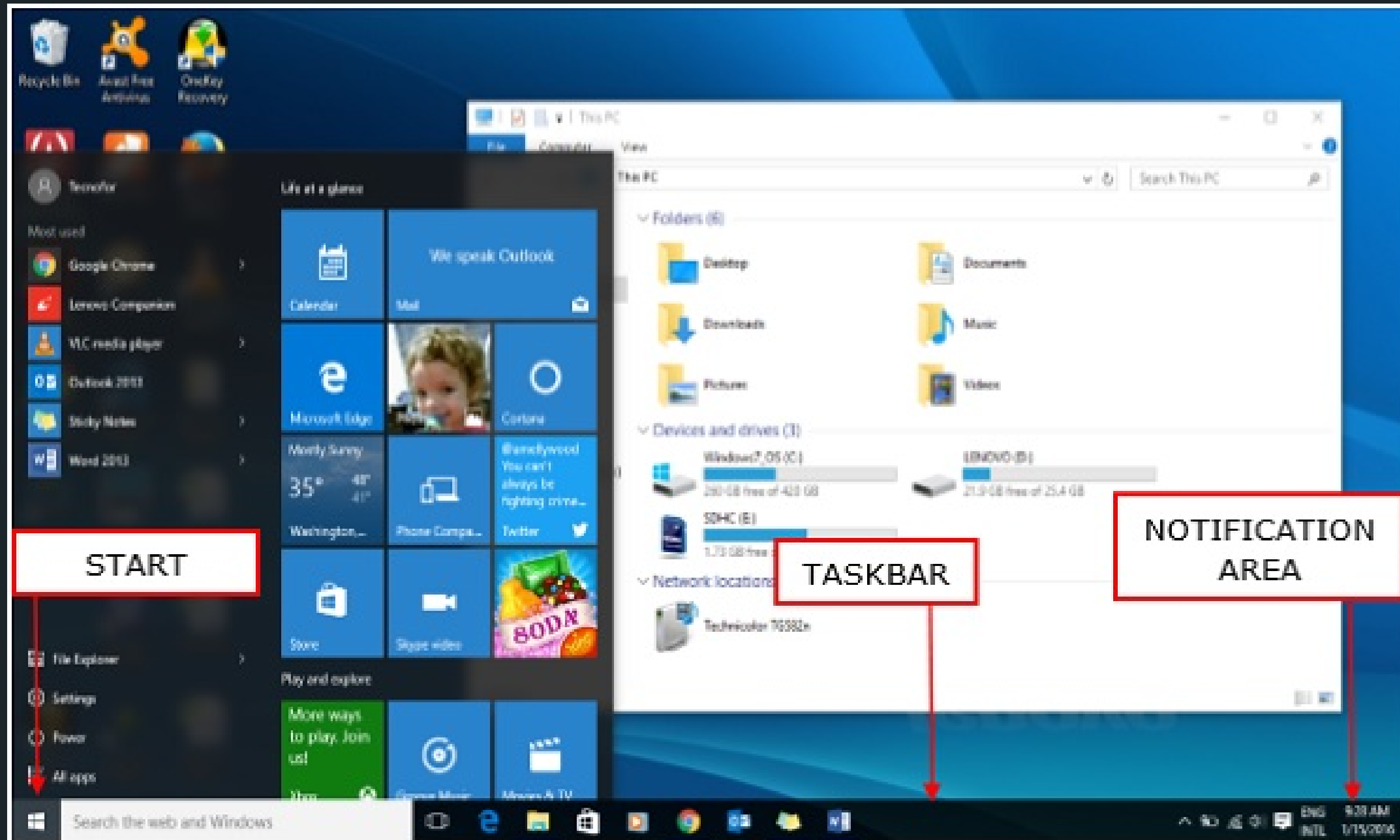
GUI AND CLI

The operating system enables the interaction between the user and the device. It does this by providing two user interfaces: **a command-line interface (CLI)** and **a graphical user interface (GUI)**.

Users of GUI interact with devices or software through clickable graphical icons. CLI is a text-based or console representation whereby users type commands into a terminal to navigate or operate devices or software.



WINDOWS GUI



MACOS GUI



LINUX GUI

